Proportional Reasoning-1 – Homework Questions

CBSE Class 8 Mathematics · Ganita Prakash (Part 1) · Chapter 7

Name: _____ Class/Section: _____ Date: _____

Instructions: Answer all 30 questions. Show your working. Write every ratio in its simplest form unless asked otherwise.

Section A – Ratios

Q1. Look at the shapes below.
(a) Write the ratio of the number of circles to the number of triangles in its simplest form.
(b) Write the ratio of the number of triangles to the total number of shapes in its simplest form.



Q2. Write each ratio in its simplest form:
(a) 12 : 18 (b) 25 : 100 (c) 14 : 21 (d) 45 : 30

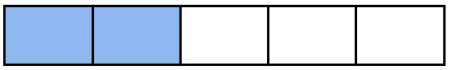
Q3. A class has 18 girls and 12 boys. Find the ratio of:
(a) girls to boys (b) boys to the total number of students

Q4. Find the ratio of 50 paise to ₹2. (Hint: first write both amounts in the same unit.)

Q5. Find the ratio of 400 g to 2 kg in its simplest form.

Q6. Fill in the blanks so that all the ratios are equal:
2 : 5 = 6 : ____ = ____ : 20 = 10 : ____

Q7. The strip below is divided into equal parts. Write the ratio of:
(a) shaded parts to unshaded parts (b) shaded parts to all the parts



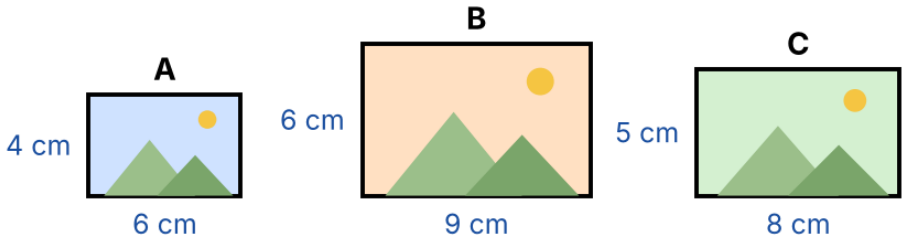
Section B – Proportion and Resizing Pictures

Q8. Are the following ratios in proportion? Write Yes or No.
(a) 3 : 4 and 9 : 12 (b) 2 : 5 and 4 : 9 (c) 6 : 8 and 15 : 20

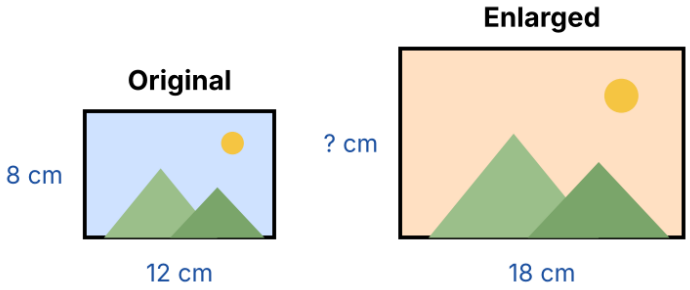
Q9. Find the missing number:
(a) 3 : 5 = x : 20 (b) 4 : x = 12 : 27

Q10. Check whether the numbers 25, 40, 30, 48 are in proportion (that is, whether 25 : 40 = 30 : 48).

Q11. Three pictures A, B and C are shown below with their widths and heights. Which two pictures have their width and height in the same ratio?

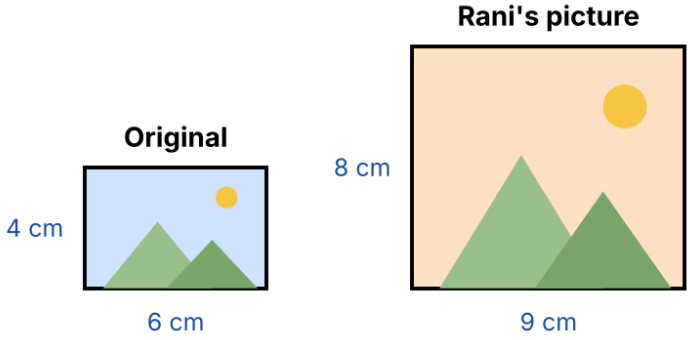


Q12. A photo is 12 cm wide and 8 cm high. It is enlarged to a width of 18 cm without changing the ratio of width to height. What is the height of the enlarged photo?



Q13. A poster is 20 cm wide and 15 cm high. It is made smaller so that its width becomes 8 cm, keeping the same ratio of width to height. What will its new height be?

Q14. Rani resized a picture that was 6 cm wide and 4 cm high into a picture 9 cm wide and 8 cm high.
(a) Is the ratio of width to height the same in both pictures?
(b) Will Rani's picture look stretched?
(c) What height should she have used with the 9 cm width?



Section C – Rule of Three

Q15. 4 pens cost ₹60. What is the cost of 7 such pens?

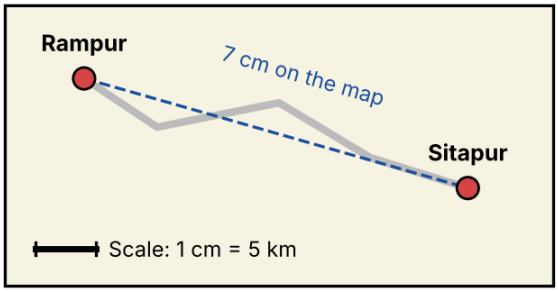
Q16. 5 kg of rice costs ₹300. How much will 8 kg of the same rice cost?

Q17. A car travels 120 km in 3 hours. At the same speed, how far will it travel in 5 hours?

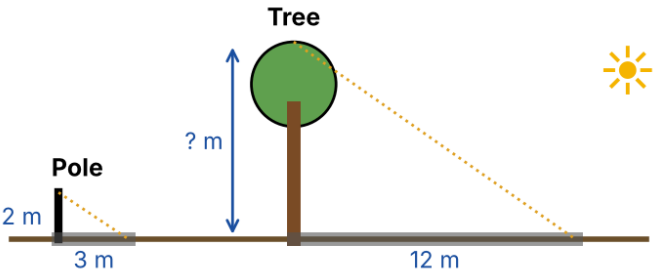
Q18. 12 notebooks cost ₹480. How many such notebooks can be bought for ₹200?

Q19. A recipe uses 2 cups of rice to serve 6 people. How many cups of rice are needed to serve 15 people?

Q20. On the map below, 1 cm stands for 5 km. The two towns are 7 cm apart on the map. What is the actual distance between Rampur and Sitapur?



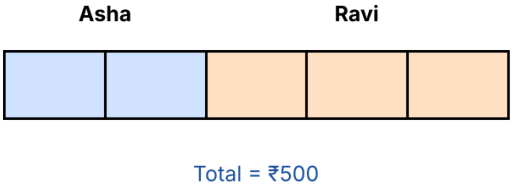
Q21. A pole 2 m tall casts a shadow 3 m long. At the same time, a tree casts a shadow 12 m long. How tall is the tree?



Q22. A machine fills 150 bottles in 5 minutes. How many bottles will it fill in 12 minutes, working at the same rate?

Section D – Sharing, but Not Equally

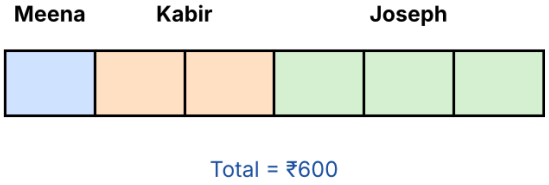
Q23. Asha and Ravi share ₹500 in the ratio 2 : 3, as shown in the bar model. How much money does each get?



Q24. 35 sweets are divided between Tina and Sonu in the ratio 3 : 4. How many sweets does each get?

Q25. In a school club, the ratio of boys to girls is 4 : 5. If there are 20 boys, how many girls are there? How many members are there in all?

Q26. Meena, Kabir and Joseph share ₹600 in the ratio 1 : 2 : 3. How much does each person get?



Total = ₹600

Q27. To make green paint, blue and yellow paint are mixed in the ratio 2 : 3. How many litres of blue paint and of yellow paint are needed to make 20 litres of green paint?

Section E – Unit Conversion

Q28. 1 foot = 12 inches. Find:
(a) 4 feet = ____ inches (b) 60 inches = ____ feet

Q29. Suppose 1 US dollar = ₹84. How many rupees is 5 US dollars?

Q30. Take 1 inch = 2.5 cm. A TV screen measures 40 inches. What is this length in centimetres?